Dr Rushani Wijesuriya

BSc (Hons), PhD · Senior Biostatistician, Murdoch Children's Research Institute
rushani.wijesuriya@mcri.edu.au · ORCID 0000-0003-1023-4065 · Google Scholar

EDUCATION

2018–2021 **PhD in Biostatistics**, The University of Melbourne
Thesis: Evaluation of multiple imputation approaches for dealing with incomplete three-level data

2012–2017 **BSc (Hons) in Statistics, First Class**, University of Colombo, Sri Lanka
GPA 3.68/4.00

EMPLOYMENT

2025– **Senior Research Officer (Biostatistician)**, Clinical Epidemiology & Biostatistics Unit, Murdoch Children's Research Institute

2021–2025 **Research Officer (Biostatistician)**, Clinical Epidemiology & Biostatistics Unit, Murdoch Children's Research Institute

2021–2022 **Biostatistician**, Department of General Practice, The University of Melbourne

2021 **Biostatistician**, Methods and Implementation Support for Clinical and Health Research (MISCH), The University of Melbourne

2018–2020 **Academic Tutor**, Centre for Epidemiology and Biostatistics, The University of Melbourne

2017–2018 **Assistant Lecturer**, Department of Statistics, University of Colombo, Sri Lanka

GRANTS AND AWARDS

2026 **NHMRC Ideas Grant**, Chief Investigator (AUD 3,100,000)

2026 **Royal Children's Hospital Foundation Grant**, Chief Investigator (AUD 806,000)

2025 **Murdoch Children's Research Institute Rising Star Award**

2024 **MCRI Population Health Theme Funding**, Principal Investigator (AUD 25,000)

2019–2023 **Travel and development awards:** ViCBiostat Professional Development Award (2023); MCRI Population Health travel grant (2023); MCRI Henry Ackman Travel Award (2019); Statistical Society of Australia travel award, Young Statisticians' Conference (2019)

2020 **MCRI Three Minute Thesis competition**, second runner-up

2018 **Melbourne Research Scholarship**

PUBLICATIONS

Methods research

1. **Wijesuriya R**, Hughes RA, Carlin JB, Peters RL, Koplin JJ, Moreno-Betancur M. A roadmap for systematic identification and analysis of multiple biases in causal inference. American Journal of Epidemiology. 2026;195(11):2802.
2. Moreno-Betancur M, **Wijesuriya R**, Carlin JB. The ideal trial: defining causal estimands that balance relevance and feasibility in target trial emulations and actual randomized trials. Epidemiology. 2026;37(2):153–162.
3. Moreno-Betancur M, **Wijesuriya R**, Carlin JB. Reflections on frameworks for defining causal estimands in both observational studies and trials: response to Labrecque et al. Epidemiology. 2026.
4. **Wijesuriya R**, Moreno-Betancur M, Carlin JB, White IR, Quartagno M, Lee KJ. Multiple imputation for longitudinal data: a tutorial. Statistics in Medicine. 2025;44(3–4):e10274.
5. **Wijesuriya R**, Moreno-Betancur M, Carlin JB, De Silva AP, Lee KJ. Multiple imputation approaches for handling incomplete three-level data with time-varying cluster memberships. Statistics in Medicine. 2022;41(22):4385–4402.
6. **Wijesuriya R**, Moreno-Betancur M, Carlin JB, De Silva AP, Lee KJ. Evaluation of approaches for accommodating interactions and non-linear terms in multiple imputation of incomplete three-level data. Biometrical Journal. 2022;64(8):1404–1425.
7. **Wijesuriya R**, Moreno-Betancur M, Carlin JB, Lee KJ. Evaluation of approaches for multiple imputation of three-level data. BMC Medical Research Methodology. 2020;20(1):207.

Health research (selected)

1. Koplin JJ, Shifti DM, Soriano VX, **Wijesuriya R**, et al. Egg allergy prevalence before and after guidelines for earlier egg introduction. JAMA Pediatrics. 2026;180(9):1015–1023.

2. Buckingham P, Tollit MA, Wong G, **Wijesuriya R**, Hoq M, Pace CC, Pang KC. Social transition in young trans people accessing gender-affirming care. International Journal of Transgender Health. 2025:1–19.
3. Priest N, Guo S, **Wijesuriya R**, et al. To what extent could eliminating racial discrimination reduce inequities in mental health and sleep problems among Aboriginal and Torres Strait Islander children? A causal mediation study. The Lancet Regional Health – Western Pacific. 2024;51:101196.
4. Guo S, **Wijesuriya R**, O'Connor M, Moreno-Betancur M, Goldfeld S, Burgner D, Liu R, Priest N. The effects of adverse and positive experiences on cardiovascular health in Australian children. International Journal of Cardiology. 2024;411:132262.
5. Parker KM, Dang TD, **Wijesuriya R**, et al. Longitudinal peanut and Ara h 2 specific-IgE, -IgG4, and -IgG4/-IgE ratios are associated with the natural resolution of peanut allergy in childhood. Allergy. 2024;79(7):1868–1880.
6. Peters RL, Soriano VX, Lycett K, Flynn C, Idrose NS, Tang MLK, **Wijesuriya R**, et al. Infant food allergy phenotypes and association with lung function deficits and asthma at age 6 years: a population-based, prospective cohort study in Australia. The Lancet Child & Adolescent Health. 2023;7(9):636–647.
7. Goh AM, Dang C, **Wijesuriya R**, et al. The impact of strict lockdowns on the mental health and well-being of people living in Australia during the first year of the COVID-19 pandemic. BJPsych Open. 2023;9(3):e90.

PRESENTATIONS

Invited

2026 — Australasian Epidemiological Association Methods Seminar Series (online)

2025 — Handling missing data in longitudinal studies using multiple imputation
MCRI LifeCourse Cohort Forum (online)

2025 — Multiple imputation: an overview
Longitudinal Cohort Study on the Filipino Child (online)

2024 — A roadmap for identifying and analysing multiple biases in causal inference
International Statistics Conference, Sri Lanka

2022 — Multiple imputation for incomplete data with multiple levels of clustering: issues and guidance
Research Matters seminar series, Department of General Practice, The University of Melbourne

2020 — Accommodating interactions and non-linear terms in multiple imputation of incomplete three-level data
International Missing Data, Imputation and Analysis (MiDIA) Group, UK

2019 — Multiple imputation in three-level data structures
Young Statisticians Showcase, Victoria

Conference oral presentations

2026 — Practical multiple bias analysis using multiple imputation
Australasian Epidemiological Association Scientific Meeting, Fiji

2025 — Simultaneous assessment of multiple biases in causal inference
Australasian Epidemiological Association Scientific Meeting, Hobart

2023 — Systematic identification and analysis of multiple biases in causal inference
International Society for Clinical Biostatistics (Milan); Australasian Epidemiological Association (Melbourne); Australian Statistical Conference (Wollongong)

2020–2021 — Multiple imputation of incomplete three-level data, including time-varying cluster memberships, interactions and non-linear terms
International Society for Clinical Biostatistics (2020, 2021, virtual); Australian and New Zealand Statistical Conference (2021)

2019 — Multiple imputation in three-level data structures
Young Statisticians Conference, Canberra; MCRI Department of Paediatrics Student Symposium, Melbourne

2017 — Comparing the performance of randomization tests and traditional tests
International Conference on Computational Modelling and Simulations, Colombo

TEACHING

Short courses and workshops

2022– — **Observational studies: modern concepts and analytic methods**
Short course, Clinical Epidemiology & Biostatistics Unit, MCRI

2023–2025 **ViCBiostat Summer School**
Advanced causal inference (2025); Causal mediation analysis (2024); Causal analysis methods beyond standard regression (2023)

2023 Causal analysis methods beyond standard regression
Workshop, Australasian Epidemiological Association Meeting

2019–2026 Multiple imputation workshops
National Allergy Centre for Research (2026); International Agency for Research on Cancer (2022); ViCBiostat (2019–2020)

2018–2020 R and Stata practicals; Introduction to Biostatistics
Short courses, Clinical Epidemiology & Biostatistics Unit, MCRI

University teaching

2018–2020 Master of Public Health, The University of Melbourne
Tutor, demonstrator and assessor: Inference Methods for Biostatistics; Biostatistics; Linear and Logistic Regression

2017–2018 University of Colombo, Sri Lanka
Lecturer and course coordinator, Essential Mathematics for Statistics; demonstrator and assessor, Fundamentals of Statistics

RESEARCH SUPERVISION

Current PhD supervisor: Qianying Tang (Biostatistics, 2026–); Sophie Rosser (Epidemiology, 2023–)
PhD advisor: Ping Tang (Epidemiology, 2025–)

Completed Master of Biostatistics research projects, supervisor: Zachary Anderson-Cutting (2025); Ming Wang (2023); Xuelu Sun (2022)
PhD advisor: Shakira Milton (Implementation Science, 2024)

LEADERSHIP AND SERVICE

- Coordinator, MCRI campus-wide data science journal club (2025–)
- Member, International Society for Clinical Biostatistics Student Award subcommittee (2023–)
- Chair, monthly Causal team meetings, Clinical Epidemiology & Biostatistics Unit (2022–2024)
- Founding member, Statistical Society of Australia Equity, Diversity & Inclusion Committee (2021–2023); co-author of the Society's 2022 equity, diversity and inclusivity survey report
- Chair, code of conduct response team, and session co-chair, Australian Statistical Conference, Wollongong (2023)
- Manager, Clinical Epidemiology & Biostatistics Unit social media accounts (2022–)
- Member, MCRI research students' association (2019)

Mentoring and outreach

- Mentor: Statistical Society of Australia Mentoring Program (2024); ScholarX, Sustainable Education Foundation, Sri Lanka (2022)
- Invited speaker on careers in biostatistics and women in STEM, Australia's Rising Women in Data Science (WiDS) Stars, Australian Data Science Network (2023)
- Panel member, Women in Data Science conference, Sri Lanka (2023)
- Career talks at International Society for Clinical Biostatistics Early Career Biostatisticians' Days, Milan (2023) and Krakow (2020)
- Co-author, "Women of Significance?", Significance. 2022;19(3):16–20

Peer review

- Journals: Statistics in Medicine; Biometrical Journal; Journal of the Royal Statistical Society: Series A; Paediatric and Perinatal Epidemiology; The Medical Journal of Australia; Journal of Educational and Behavioral Statistics
- In-house biostatistics reviewer, Journal of Allergy and Clinical Immunology: In Practice
- Reviewer, International Conference in Data Science (2023); judge, MCRI student symposium (2022)

Memberships

- International Society for Clinical Biostatistics (2020–); Statistical Society of Australia (2018–); Victorian Centre for Biostatistics (2018–)